This gives a nice framework for thinking about why one particle has a different mass from another, but, as of today, there is no fundamental explanation for the precise manner in which each of the known particle species interacts with the Higgs ocean. As a result, there is no fundamental explanation for why the known particles have the particular masses that have been revealed experimentally. However, most physicists do believe that were it not for the Higgs ocean, *all fundamental particles would be like the photon and have no mass whatsoever.* In fact, as we will now see, this may have been what things were like in the earliest moments of the universe.

Unification in a Cooling Universe

Whereas gaseous steam condenses into liquid water at 100 degrees Celsius, and liquid water freezes into solid ice at 0 degrees Celsius, theoretical studies have shown that the Higgs field condenses into a nonzero value at a million billion (10^{15}) degrees. That's almost 100 million times the temperature at the core of the sun, and it is the temperature to which the universe is believed to have dropped by about a hundredth of a billionth (10^{-11}) of a second after the big bang (ATB). Prior to 10^{-11} seconds ATB, the Higgs field fluctuated up and down but had an average value of zero; as with water above 100 degrees Celsius, at such temperatures a Higgs ocean couldn't form because it was too hot. The ocean would have evaporated immediately. And without a Higgs ocean there was no resistance to particles undergoing accelerated motion (the paparazzi vanished), which implies that all the known particles (electrons, up-quarks, down-quarks, and the rest) had the same mass: zero.

This observation partly explains why the formation of the Higgs ocean is described as a cosmological phase transition. In the phase transitions from steam to water and from water to ice, two essential things happen. There is a significant qualitative change in appearance, and the phase transition is accompanied by a reduction in symmetry. We see the same two features in the formation of the Higgs ocean. First, there was a significant qualitative change: particle species that had been massless suddenly acquired nonzero masses—the masses that those particle